STUDENT-SUBMITTED QUESTIONS & ANSWERS
EXAM II REVIEW DAY

ELEMENTARY LINEAR ALGEBRA (MATH 2250-03)

SPRING, 2019

INTRODUCTION

In this document, I have compiled all of the questions that were sent to me in preparation for the Exam 2 Review Day. These questions are near-verbatim the questions I was sent (minus any personally identifying information and with minor formatting changes to make the math display nicely). After each question, I've provided a short answer. Hopefully, seeing the questions your classmates asked (phrased exactly as they asked them) will provide a helpful 'on-the-ground' perspective regarding the types of questions you might have and the answers to them.

QUESTIONS

1. In regards to question 1 from homework 6: What does it mean to be 'closed' under addition and scalar multiplication in order to satisfy:

   $$(1)\ \ 0 \in \mathrm{H}, \quad (2)\ \ \mathrm{H} \text{ is closed under addition}, \quad (3)\ \ \mathrm{H} \text{ is closed under scalar multiplication}.$$

   Does it simply mean that by plugging in 0, the result is 0, like the solution for this question implies ($(p_1 + p_2)(0) = 0$, $(kp)(0) = 0$)? If so, then in what case would this NOT be true? Or am I simply misunderstanding the definition of being closed under these conditions?
   Question 1 on homework 6 asks:

   > *Let* V *be the set of all polynomials in* $\mathbb{P}_n$ *such that* $p(0) = 0$, *i.e.*
   >
   > $$\mathrm{V} = \left\{ p \in \mathbb{P}_n : p(0) = 0 \right\}.$$
   >
   > *Show that* V *is a subspace of* $\mathbb{P}_n$.

   I think some of the confusion your having is related to the wording and language used in the key when I answered this problem. Namely, the first part of the answer to this problem (as given in the key) reads:

   > *Recall that a subspace of a vector space* W *is a subset* H *of* W *such that:*
   >
   > $$(1)\ \ 0 \in \mathrm{H}, \quad (2)\ \ \mathrm{H} \textit{ is closed under addition}, \quad (3)\ \ \mathrm{H} \textit{ is closed under scalar multiplication}.$$
   >
   > *So, to show that* V *is a subspace of* $\mathbb{P}_n$, *we need only show that* V *satisfies the three requirements above.*

   So, the conditions you've listed aren't just the requirements for *this particular* subset of $\mathbb{P}_n$ to be a subspace of $\mathbb{P}_n$ – they're the requirements *any* subset W of *any* vector space V needs to satisfy to be deemed a subspace. This may seem like kind of a nit-picky distinction, but thinking like this is very helpful when forming intuition for the answer to this type of question. That is, since this is a *general* set of conditions need for *any* subset of an *arbitrary* vector space to be a subspace, the conditions given therein *aren't* dependent on the specifics of this problem. So, it's not that "closure under addition" means *"that by plugging in 0, the result is 0."* Instead, that's the way that this *general condition* appears in this *specific case*.

Let's be more precise: Saying a set V is "closed under addition" is a short-hand way of saying "if $u \in V$ and $v \in V$, then $u + v \in V$." In this question, our set V is the set

$$V = \Big\{p \in \mathbb{P}_n : p(0) = 0\Big\}.$$

Recall that the notation used to define V is called "set builder notation" and works like this: If you want a set, let's call it A, of things belonging to a set X that satisfy some condition C, then we choose a symbol to represent the things in our set (let's choose x to be this symbol) and write

$$A = \Big\{x \in X : x \text{ satisfies condition } C\Big\}.$$

Read aloud, this says "A equals the set of x in X such that x satisfies the condition C."

So, for this problem, the set-builder notation used in defining V says that "V is the set of polynomials p in $\mathbb{P}_n$ such that p satisfies the condition $p(0) = 0$." So, since V being "closed under addition" means that having $p, q \in V$ implies $p + q \in V$, we're just using the *definition* of V that set-builder notation gives us to determine if $p + q$ is a polynomial in V, since saying $p + q \in V$ is *the same thing* as saying $(p + q)(0) = 0$ (i.e. "plugging in zero and getting zero" is *the definition* of being in V). So, if V was defined as the set of polynomials which satisfy *some other condition*, then we'd just have to check whether or not the sum of two elements in V satisfies *that other condition* instead if we wanted to show that V was closed under addition (since having the sum of two vectors be "in" V is the same as saying that the sum of the vectors satisfies the condition which defines V).

The same line of reasoning works for scalar multiplication as well, but instead we're asking whether or not a scalar multiple of something in V is also in V. That is, if $q \in V$ and $c \in \mathbb{R}$ is a scalar, is it true that $cq \in V$? To show this, we would similarly have to show that if q is in V (that is, if q is a polynomial which satisfies the property that defines V), then $cq \in V$ (that is, cq is a polynomial which satisfies the property that defines V). If you go back and check through the solutions to problems 1 in the key for homework 6, you'll see that this is exactly the process that the answer follows for each of the conditions described above.

2. I think after attempting this last homework, I've come to understand that I'm confused as to how to break down a span for a problem. If we are given a matrix or column vectors and asked to determine if it is in fact a span for some dimension or anything else, I feel comfortable working through those types of problems. But working from the other way around, finding a span for some set of values, is what I find hard to do. What would be a step by step process that could be followed in order to simplify an expression to be the span of something?
For instance, on problem 6 from homework 6, I was able (with the help of the hint given) to simplify the expression down to $\Big\{\sin(t), 2\sin(t)\cos(t), \sin(t)\cos(t)\Big\}$, but didn't know how to proceed from there. From the solution to the problem, it appears it was broken down by inserting constants c_1, c_2, and c_3 for each of the terms while also forming an equation for the whole system so that it looked like:

$$\operatorname{span}\Big\{\sin(t), \sin(2t), \sin(t)\cos(t)\Big\} = \Big\{c_1 \sin(t) + 2c_2 \sin(t)\cos(t) + c_3 \sin(t)\cos(t) : c_1, c_2, c_3 \in \mathbb{R}\Big\}.$$

Is this the correct process to do for any span problem?
First, let's clear up some notational stuff, since being on the same page notation-wise will help us to be sure that we're talking about the same thing. For the first note, you say in your question:

"If we are given a matrix or column vectors and asked to determine if it is in fact a span for some dimension or anything else, I feel comfortable working through those types of problems."

There's one issue here, which I think is maybe sort of common. Namely, a set of vectors *never* spans "dimensions" – a set of vectors can only span a *vector space* or a *subspace* of a vector space. "Dimension" is just a *number* associated to a vector space or a subspace of a vector space which tells you the *size* of any linearly independent spanning set. If you're trying to talk about a dimension in the colloquial way, like how people say "a spatial dimension" or something like that, then the precise linear algebra term for this would be a *one dimensional subspace*. That is, when somebody casually says, for example, "this world around you has three dimensions: length, width, and height" what they mean (in precise linear algebra terminology) is something like "$\mathbb{R}^3$ is a three-dimensional vector space, so it has three one-dimensional subspaces which we'll call length, width, and height."

Also, we should be careful here: a matrix can *only* span a vector space *if the vectors in that vector space are matrices*. So, it is *never* correct to say that a matrix A spans $\mathbb{R}^n$ unless A is an $n \times 1$ matrix (that is, a column vector), since the "stuff" in any set of thing that span a vector space *must* belong *to that vector space*. So, for example, while the columns of a 3×2 matrix could span $\mathbb{R}^3$, the matrix itself can *never* span $\mathbb{R}^3$ since it *isn't in* $\mathbb{R}^3$. To contrast however, an $n \times n$ matrix *can* certainly span a subspace of $M_{n\times n}(\mathbb{R})$, since the *vectors* in $M_{n\times n}(\mathbb{R})$ are *matrices*.

That aside, let's answer the actual question: what process can we employ to systematically solve 'span problems'? To answer this question, it will be helpful to review what span means in the first place.

Given a vector space V and set of vectors $\left\{v_1, \ldots, v_p\right\}$ belonging to V (that is, a subset of V), the **span** of $\left\{v_1, \ldots, v_p\right\}$ is the set of elements of V which are linear combinations of the vectors $v_1, \ldots, v_p$. That is,

$$\operatorname{span}\left\{v_1, \ldots, v_p\right\} = \left\{x \in V : x = c_1 v_1 + \ldots + c_p v_p,\ \ c_1, \ldots, c_p \in \mathbb{R}\right\}.$$

So, in this problem, when I wrote in the answer key that

$$\operatorname{span}\left\{\sin(t), \sin(2t), \sin(t)\cos(t)\right\} = \left\{c_1 \sin(t) + 2c_2 \sin(t)\cos(t) + c_3 \sin(t)\cos(t) : c_1, c_2, c_3 \in \mathbb{R}\right\}$$

I was just writing what the span of the vectors $\left\{\sin(t), \sin(2t), \sin(t)\cos(t)\right\}$ (in the vector space of all real-valued functions) is *by the definition* of span. That is,

$$\operatorname{span}\left\{\sin(t), \sin(2t), \sin(t)\cos(t)\right\}$$

is just a short-hand name we give to the *set of vectors*

$$\left\{c_1 \sin(t) + 2c_2 \sin(t)\cos(t) + c_3 \sin(t)\cos(t) : c_1, c_2, c_3 \in \mathbb{R}\right\}.$$

i.e. We're just using the "span" terminology to save ourselves some time when writing things down. So, when you say

"From the solution to the problem, it appears it was broken down by inserting constants c_1, c_2, and c_3 for each of the terms while also forming an equation for the whole system"

you are completely correct in the sense that this is *the definition of span*, which we can/should certainly use in *any* answer to *any* question involving the span of some vectors. However, the question you're referencing is *not* a "span" problem (as you term it), it's a *basis* problem. Namely, the question asks:

In the vector space of all real-valued functions, find a basis for the subspace spanned by $\left\{\sin(t), \sin(2t), \sin(t)\cos(t)\right\}$.

So, the approach used to solve this question in the key is a technique to solve questions which ask for the *basis* of a subspace *spanned* by a given set of vectors. That is, this problem isn't about what the span of these vectors is, it's about finding a *basis* for the subspace spanned by these vectors. Since a basis for a subspace W of a vector space V is *by definition* a linearly independent set of vectors in W which spans W, this explains the technique used in the answer key to solve this problem:

(a) Since we know that our subspace is the span of the vectors $\{\sin(t), \sin(2t), \sin(t)\cos(t)\}$, we already have a set of vectors which spans our subspace.

(b) To find a basis for our subspace, we just need to find a *linearly independent set of vectors* which spans our subspace.

(c) To find a linearly independent set of vectors which spans our subspace, we just take the set of vectors *that we already know* spans our subspace and we get rid of some of them until the remaining set of vectors is linearly independent.

So, in this case, it is easy to do this (in a nationally compact format) by simply writing out the subset of vectors that $\text{span}\{\sin(t), \sin(2t), \sin(t)\cos(t)\}$ is a shorthand for, performing algebraic manipulations on the linear combination which defines this set, and re-writing it as the span of a linearly independent subset of the vectors $\sin(t), \sin(2t), \sin(t)\cos(t)$. In general, this is a good approach for any problem which asks you to find a basis for a subspace given as the span of some vectors.

3. Could you explain the difference between a basis, a subspace, and a subset? They all seem very similar to me and I'm having a hard time keeping them straight.

To answer your question, we'll describe the difference between a basis, a subspace, and a subset by giving the definitions of each of these things and highlighting the properties that vary between them with examples. We'll start with the most general objects you asked about and work our way up to the more structured objects you mentioned.

Let's start with something you didn't ask and describe what a "set" is in mathematics:

A **set** is a collection of distinct objects, considered as an object in its own right.

For example, the numbers 2, 4, and 6 are distinct objects when considered separately, but when they are considered collectively they form a single set of size three, written $\{2, 4, 6\}$.

With the idea of a set in hand, we can define a subset:

If every element (i.e. distinct object) of set A is also in B, then A is said to be a **subset** of B. This is written $A \subseteq B$, which is pronounced "A is contained in B."

For example, the set $A = \{2, 4\}$ is a subset of the set $B = \{2, 4, 6\}$, since every element belonging to A is also in B.

That is, a subset is a *type of set* which satisfies the *additional property of being contained in another set*.

Now, we're ready to talk about subspaces:

A **subspace** W of a vector space V is a subset $W \subseteq V$ which is *also* a vector space. That is, W is a collection of things in V which, when considered on its own, is a vector space.

For example, the set $W = \left\{ \begin{bmatrix} x \\ 0 \end{bmatrix} \in \mathbb{R}^2 : x \in \mathbb{R} \right\}$ is a *subset* of $\mathbb{R}^2$ (since everything in W is also in $\mathbb{R}^2$) which is *also* a vector space *by itself*. To contrast, the set $H = \left\{ \begin{bmatrix} x \\ 1 \end{bmatrix} \in \mathbb{R}^2 : x \in \mathbb{R} \right\}$ is a *subset* of $\mathbb{R}^2$ which is *not* a vector space by itself (since the zero vector isn't in H). So, H is a subset of $\mathbb{R}^2$, while W is both a *subset* and a *subspace* of $\mathbb{R}^2$.

That is, subspaces are a *type of subset* of a vector space which satisfies the *additional property of being a vector space when considered by itself*.

Finally, we're ready to talk about a basis:

A **basis** for a subspace W of a vector space V is a subset $\mathcal{B} \subseteq W$ such that $\mathcal{B}$ is linearly independent and spans W.

For example, the set $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ is a subset of $\mathbb{R}^2$ (that is, $\mathcal{B} \subseteq \mathbb{R}^2$) which is *also* a basis for the subspace W given in the previous example, since $\mathcal{B}$ is linearly independent and spans W.

That is, a basis for a subspace W of a vector space V is a *type of subset* of W which satisfies the *additional properties* of being *linearly independent* and *spanning* W.

4. My question for extra credit is on Homework 6 Section 4.4 number 7. I don't understand why the matrix $\begin{bmatrix} 2 & 1 \\ -9 & 8 \end{bmatrix}$ is the answer. Shouldn't it be multiplied by some constants to reach the value necessary to reach the standard basis? Even the explanation shows the constants. In other words, I do not follow the logic that makes the basis $\mathcal{B}$ be the change of matrix coordinates for the standard matrix. It just seems counter intuitive.
To answer this question, let's start with the definition of a coordinate vector:

Given a basis $\mathcal{B} = \{b_1, \ldots, b_n\}$ for a subspace W of $\mathbb{R}^m$ and a vector $x \in W$, the **$\mathcal{B}$-coordinate vector** of x, which we denote by $[x]_{\mathcal{B}}$, is the vector

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$$

whose entries satisfy the equation

$$x = c_1 b_1 + \ldots + c_n b_n.$$

So, since the linear combination $c_1 b_1 + \ldots + c_n b_n$ can be re-written as

$$c_1 b_1 + \ldots + c_n b_n = \begin{bmatrix} b_1 & \ldots & b_n \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots c_n \end{bmatrix} = \begin{bmatrix} b_1 & \ldots & b_n \end{bmatrix} [x]_{\mathcal{B}}$$

the matrix M which satisfies $M[x]_{\mathcal{B}} = x$ is precisely $M = \begin{bmatrix} b_1 & \ldots & b_n \end{bmatrix}$. We *define* this matrix to be the **change-of-coordinates matrix from the basis $\mathcal{B}$ to the standard basis for $\mathbb{R}^m$**. That is, the "change-of-coordinates matrix from the basis $\mathcal{B}$ to the standard basis for $\mathbb{R}^m$" is the nickname we give to the matrix M which satisfies $M[x]_{\mathcal{B}} = x$ for all $x \in \mathbb{R}^m$. As we have shown above, this matrix M is always the matrix $\begin{bmatrix} b_1 & \ldots & b_n \end{bmatrix}$. So, to answer your question, the answer to problem 7 in Section 4.4 of Homework 6 is $\begin{bmatrix} 2 & 1 \\ -9 & 8 \end{bmatrix}$ because the basis for $\mathbb{R}^2$ that we're given is $\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \end{bmatrix} \right\}$.

It may seem like a disappointment, but there really isn't any fancy trick or deeper understanding to have beyond this. Really, at the end of the day, we're just doing basically the *only* thing that *anyone ever does* in mathematics: We're showing that some specific object or class of objects satisfies some property/properties that we decided we wanted to study, then we're giving a name to the object/objects to make it/them easier to reference later on. Sometimes, mathematicians will flip the order of this process and name an object which satisfies some property before they figure out what it is (we call these "unsolved problems" in general), but its basically just the backwards version of the same thing.

5. If A is an $m \times n$ matrix that is $m \times 7$. If $\dim(\mathrm{Col}(A)) = 3$, find $\dim(\mathrm{Null}(A))$. How many possible values of m are there?
I believe that the answer is 3, as $\dim(\mathrm{Col}(A)) \leq m$ and $n = \dim(\mathrm{Nul}(A)) + \dim(\mathrm{Col}(A))$ tells us $7 = (4) + 3$, so m must be less than or equal to 3, giving three possible values, 1,2 and 3, as if $m = 0$ the matrix wouldn't exist.

Recall that the Rank theorem states that if A is an $m \times n$ matrix, then

$$\dim\big(\mathrm{Col}(A)\big) + \dim\big(\mathrm{Nul}(A)\big) = n.$$

If I'm understanding this question correctly, A is an $m \times 7$ matrix and $\dim\big(\mathrm{Col}(A)\big) = 3$. So,the Rank theorem then tells us that

$$\begin{aligned} \dim\big(\mathrm{Col}(A)\big) + \dim\big(\mathrm{Nul}(A)\big) &= n \\ 3 + \dim\big(\mathrm{Nul}(A)\big) &= 7 \\ \dim\big(\mathrm{Nul}(A)\big) &= 7 - 3 = 4. \end{aligned}$$

That is, there are *not* multiple possible values for $\dim\big(\mathrm{Nul}(A)\big)$–there's *only one possible value*.

However, if we instead consider a general $m \times 7$ matrix *and we don't impose the the condition that* $\dim\big(\mathrm{Col}(A)\big) = 3$, there can certainly be multiple possible values for $\dim\big(\mathrm{Nul}(A)\big)$. Namely, since an $m \times 7$ matrix has m rows, A can have at most m pivots (one per row). So, as $\dim\big(\mathrm{Col}(A)\big)$ is equal to (among other things) the number of pivots in A, we know that $0 \le \dim\big(\mathrm{Col}(A)\big) \le m$. Now, notice that the Rank theorem tells us

$$\begin{aligned} \dim\big(\mathrm{Col}(A)\big) + \dim\big(\mathrm{Nul}(A)\big) &= n \\ \dim\big(\mathrm{Col}(A)\big) + \dim\big(\mathrm{Nul}(A)\big) &= 7 \\ \dim\big(\mathrm{Nul}(A)\big) &= 7 - \dim\big(\mathrm{Col}(A)\big). \end{aligned}$$

Therefore, as the dimension of Col(A) is between 0 and m, we have

$$7 - m \le \dim\big(\mathrm{Nul}(A)\big) \le 7.$$

That is, there are exactly $7 - (7 - m) = m$ possible values for $\dim\big(\mathrm{Nul}(A)\big)$.

6. I would appreciate a review of the relationship between the "coordinates matrix" and a "basis". A specific example would be:

   (Note: This is question 7 from Homework 6) *The set* $\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \end{bmatrix} \right\}$ *is a basis for* $\mathbb{R}^2$. *Find the change-of-coordinates matrix from the basis* $\mathcal{B}$ *to the standard basis of* $\mathbb{R}^n$.

   Also, even though none of the questions referenced on the study guide involve it, I feel my understanding of the "kernel" is subpar. If we have time, I would greatly appreciate it if you could refresh us on what the "kernel" is with regard to Question 3 on HW6.
   For the relationship between a coordinate vector, a change-of-basis matrix, and a basis in the context of the quoted question, see the answer to question 4 above.

   For the question about the kernel of a transformation, you can think of the kernel as a generalization of the null space of a matrix. Namely, given a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$, the kernel of T is just the stuff in $\mathbb{R}^n$ which T sends to the zero vector in $\mathbb{R}^m$, i.e. the kernel is just the set

   $$\ker(T) = \big\{x \in \mathbb{R}^n : T(x) = 0\big\}.$$

   In fact, if we let A be the $m \times n$ standard matrix of T (i.e. the matrix A such that $T(x) = Ax$ for all $x \in \mathbb{R}^n$), then we have

   $$\ker(T) = \mathrm{Nul}(A).$$

   Basically, for linear transformations, the kernel is just a way to talk about the null space of a matrix without having to know *what that matrix is*. Instead, all we need to know is what the matrix *does* – that is, if you start with $x \in \mathbb{R}^n$, all you need to know is what the vector $b \in \mathbb{R}^m$ that T sends x to.

This is particularly useful in infinite dimensional vector spaces, or even just in finite-dimensional vector spaces which don't "look like" $\mathbb{R}^n$ right out of the box. For example, the derivative is a linear operator on $\mathbb{P}_n$. But do you know what the matrix for the derivative is in this case? Probably not, but you almost definitely know what the derivative *does* to a polynomial. So, you can "skip the middleman" so to speak and jump directly to calculating the kernel. And, like magic, you've calculated the null space for the matrix of the derivative (as a linear transformation on $\mathbb{P}_n$) *without ever knowing what that matrix is*. It's a good trick!

More precisely, what the kernel measures is, essentially, the degree to which a linear transformation *fails to be injective*. That is, if $T : X \to Y$ is a linear transformation between the vector space X and the vector space Y, then if the equation $T(x) = b$ has a solution $x = a$ (i.e. so, a vector $a \in X$ such that $T(a) = b$), *all* of the solutions to the equation $T(x) = b$ are given by $x = a + w$, where $w \in \ker(T)$. This follows from the linearity of T: If w belongs to the kernel of T, then $T(x + w) = T(x) + T(w) = T(x) + 0 = T(x)$ for every $x \in X$. So, by measuring the *dimension* of the subspace ker(T) of X which T sends to the zero vector in Y, we are *automatically* measuring the just how much "stuff" in X the transformation T is sending to any point $b \in Y$ for which $T(x) = b$ has a solution.

To see how this works out in the context of Problem 3 in Homework 6, see Question 10 in this document.

7. The question I found is a two-part, Section 4.5 Question 21 and 23. The first question (21) seems to be a good review of finding a basis and 23 is just a double check on change of coordinate. It is like the final question in homework 6 but done in a different order.

   The question and answer provided by the student is given below:

   The first four Hermite polynomials are 1, $2t$, $-2+4t^2$ *and* $-12t+8t^3$ *Show the first four Hermite polynomials form a basis of* $\mathbb{P}_3$ *which is* $\mathcal{B}$ *and Let* $p(t) = 7-12t-8t^2+12t^3$ *and find the coordinate vector of p relative to* $\mathcal{B}$.

   The first part (question 21) is relatively simple. With a standard basis of $\left\{1, t, t^2, t^3\right\}$

$$\begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 2 & 0 & -12 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

   this then row reduces to

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

   so it does form a basis (linear independent).

   As for the second part (question 23), Is this simply doing a summation of the first four Hermite polynomials with a coefficient in front of each, setting that equal to $p(t)$ and solving for the coefficients?
   The student who sent this question is totally right for the first part of this question: putting the Hermite polynomials into the standard basis, forming a matrix from the resulting vectors, and row-reducing this matrix to the identity is certainly a good way to show that these vectors are linearly independent and span the space $\mathbb{P}_3$.

   For the second part of this question, there's a slightly faster way to find the coordinate vector relative to $\mathcal{B}$ for the polynomial $p(t)$ given in the question. Namely, if $\mathcal{E}$ is the standard basis for $\mathbb{P}_3$, then the change-of-coordinates formula we talked about in section 4.7 tells us that

$$[p(t)]_{\mathcal{E}} = [\mathcal{B}]_{\mathcal{E}}[p(t)]_{\mathcal{B}}.$$

So, we can compute $[p(t)]_{\mathcal{B}}$ by computing $[\mathcal{B}]_{\mathcal{E}}^{-1}[p(t)]_{\mathcal{E}} = [p(t)]_{\mathcal{B}}$. Specifically, as you already computed,

$$[\mathcal{B}]_{\mathcal{E}} = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 2 & 0 & -12 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

and, in the standard basis, we have $[p(t)]_{\mathcal{E}} = \begin{bmatrix} 7 \\ -12 \\ -8 \\ 12 \end{bmatrix}$. So, we get

$$[p(t)]_{\mathcal{B}} = [\mathcal{B}]_{\mathcal{E}}^{-1}[p(t)]_{\mathcal{E}} = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 2 & 0 & -12 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}^{-1} \begin{bmatrix} 7 \\ -12 \\ -8 \\ 12 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ -2 \\ 3/2 \end{bmatrix}.$$

8. Let me know if this question is too broad. But I am still confused about infinite dimensional vector spaces. So, my question is if you can give a test like example using an infinite dimensional vector space?

ANNOUNCEMENT: MATERIAL ON INFINITE DIMENSIONAL VECTOR SPACES WILL NOT BE ON THIS TEST. THIS ANSWER IS GIVEN PURELY FOR NON-STUDY-RELATED PURPOSES.

Understanding infinite-dimensional vector spaces is generally a little bit tricky. On the surface level, it seems easy: Since the dimension of a vector space V is the number of elements in any basis $\mathcal{B}$ of V, any vector space which *doesn't have a finite basis* is, by definition, infinite dimensional.

For an example of this, take the vector space $\mathbb{P}(\mathbb{R})$ of all polynomials with real coefficients. It's easy to see that this space doesn't have a finite basis: Just suppose the set $\mathcal{B} = \left\{p_1(t), \ldots, p_n(t)\right\}$ is a basis for $\mathbb{P}(\mathbb{R})$ and let $p_r(t)$ be the polynomial with highest degree in $\mathcal{B}$ (if a couple of the polynomials in $\mathcal{B}$ are tied for the highest degree, just pick on at random). Then, we can take the polynomial $q(t) = t\left(p_r(t)\right)$. Since $p(t)$ is a polynomial of a higher degree than any of the elements of $\mathcal{B}$, it certainly isn't in the span of $\mathcal{B}$. But, it's a polynomial with real coefficients, so $q(t) \in \mathbb{P}(\mathbb{R})$! Hence, $\mathbb{P}(\mathbb{R})$ can't have a finite basis, since assuming it does leads to a contradiction.

But after looking at simple examples like this, the problems start. For one, a basis for a vector space V is a linearly independent set of vectors in V which spans V, but the definitions of "span" and "linearly independent" *both rely on the concept of a linear combination*. So, since a linear combination is, *by definition* a finite sums of scalar multiples of a set of vectors, it isn't *at all clear* what "linear combinations" mean in this context of infinite dimensional vector spaces! We would obviously need to define what a linear combination should mean in this context if we want to figure out what an infinite dimensional vector space is, at least. This is where a lot of sort-of-technical trickery regarding *different types of bases* comes into play, with the main two candidates being "Schauder bases" and "Hammel bases" (both named after some mathematicians who wrote about this kind of stuff). We've been working with Hammel bases in this class, though we haven't had a reason to mention that there's a distinction. If you want to learn more about Schauder bases (the type of basis you'll need to talk about an infinite dimensional vector space), I highly recommend the *truly excellent* functional analysis textbook (and believe me, I've read a few functional analysis textbooks) titled "A Course in

Functional Analysis" by John B. Conway (who, by the way, is a phenomenal mathematican). Functional Analysis is, broadly speaking, the study of most of the "nice" infinite-dimensional vector spaces, and it's one of the most coolest, most fun branches of mathematics out there. Check it out!

9. My question is under the Homework 6 material I think.

   Is the basis of a subspace for a set of vectors always a linear independent set. If not when is there an exception to this rule. In addition, can you explain why this is a thing? Why does the basis for a subspace a linear independent set.
   Yes, *by definition* a basis for a subspace W of a vector space V is a linear independent set of vectors belonging to W which spans W. There are no exceptions to this rule–we *defined* a basis this way! So, saying that

   $$\text{"}\left\{v_1,\ldots,v_p\right\} \textit{ is a basis for the subspace } W \textit{ of the vector space } V\text{"}$$

   is always *exactly the same* as saying

   $$\text{"}\left\{v_1,\ldots,v_p\right\} \textit{ is a linearly independent set of vectors belonging to the subspace } W \textit{ of the vector space } V \textit{ which spans } W.\text{"}$$

   Think of things like this as acronyms or abbreviations for linear algebra concepts. For example, the acronym "NASA" with regard to space agencies always means "National Aeronautics and Space Administration." There aren't exceptions to this rule, nor is this really a "rule" at all–it's just a shorter way of saying something!

   Now, the second part of this question is the thing that you *really* want to ask, which is "Why did we bother to come up with a nickname for this type of set of vectors in the first place?" It's a really good question–after all, *shouldn't* our first instinct be to ask *why* anyone should even bother with studying these seemingly-arbitrary sets of vectors? Why not choose some other thing to study instead? What makes these things special or useful or important at all? This kind of question really gets at the true essence of mathematics (imo, at least). I mean, there's a truly, bizarrely, *incomprehensibly* gigantic number of mathematical objects we could study, so why this one?

   The answer is, in my opinion, not terrifically exciting: we study bases *because they make vector spaces easier to work with*, and vector spaces are *everywhere* (you're living in one right now, even). Mainly, a basis let's you understand how a linear transformation $T : X \to Y$ between a vector space X and a vector space Y (which, in general, each consist of *an infinite number of vectors*) works by just understanding what T does to a basis for X (which, in general, consists of a *much smaller number of things* and, in the cases we study in this class, a *finite number of things*). The advantage here is obvious: nobody has time to see what a transformation does to every vector in a finite dimensional vector space (heck, there's never enough time for that), but seeing what a transformation does to a finite number of things (for example, a basis) doesn't take too long. This works because a basis $\mathcal{B} = \left\{b_1,\ldots,b_n\right\}$ for a finite-dimensional vector space X lets us write any vector x belonging to X as a linear combination of the elements of $\mathcal{B}$, i.e. as $x = c_1 b_1 + \ldots + c_n b_n$ for some scalars $c_1,\ldots,c_n$. This is good, because a linear transformation T can be broken apart along *precisely this type of combination of vectors*. That is, since T is linear,

   $$T(x) = T\left(c_1 b_1 + \ldots + c_n b_n\right) = c_1 T(b_1) + \ldots + c_n T(b_n).$$

   Because this decomposition works for *every vector in* X, this means that by knowing what the vectors $T(b_1),\ldots,T(b_n)$ are, we know what $T(x)$ is for *every vector x belonging to* X.

10. From HW 6, Question 3:

I understand how to show that T(A) is a linear transformation, and I can see how you used a "generic" 2x2 matrix to further visualize $A = A^T$, but can you further clarify how you were able to determine the values of x and then a,d,b,and c to define ker(T)? More sepcifically, why can you determine $a = d = 0$ and $b = -c$, $c = -b$...

Here's Question 3 from Homework 6:

Let $M_{2\times 2}(\mathbb{R})$ be the vector space of all 2×2 matrices over $\mathbb{R}$, and define $T : M_{2\times 2}(\mathbb{R}) \to M_{2\times 2}(\mathbb{R})$ by $T(A) = A + A^T$. Show that T is a linear transformation and give a description of the kernel of T.

Here's the answer that I gave in the key:

To show that T is a linear transformation, we need only show that $T(A + B) = T(A) + T(B)$ and $T(cA) = cT(A)$ for all $A, B \in M_{2\times 2}(\mathbb{R})$ and all $c \in \mathbb{R}$. We may show that $T(A + B) = T(A) + T(B)$ with the following calculation:

$$\begin{aligned} T(A+B) &= (A+B) + (A+B)^T \\ &= A + B + A^T + B^T \\ &= \left(A + A^T\right) + \left(B + B^T\right) \\ &= T(A) + T(B). \end{aligned}$$

Similarly, we may show $T(cA) = cT(A)$ by noting

$$T(cA) = (cA) + (cA)^T = c\left(A + A^T\right) = cT(A).$$

So, T is a linear transformation.

To find the kernel of T, recall that

$$\ker(T) = \left\{A \in M_{2\times 2}(\mathbb{R}) : T(A) = 0\right\}.$$

So, we need only find the elements of $M_{2\times 2}(\mathbb{R})$ which T maps to the zero matrix. To do so, observe that if $T(A) = 0$, then $A + A^T = 0$. Rearranging, this may be stated as $A = -A^T$. A matrix which satisfies this property is known as a *skew-symmetric matrix*, and may be characterized in the 2×2 case by the following observation:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = -\begin{bmatrix} a & b \\ c & d \end{bmatrix}^T$$
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} -a & -c \\ -b & -d \end{bmatrix}.$$

So, since the only real number x which satisfies $x = -x$ is $x = 0$, we may see that $a = d = 0$. Similarly, we may see from the above that we must have both $c = -b$ and $b = -c$. So, we may conclude that

$$\ker(T) = \left\{\begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix} : b \in \mathbb{R}\right\}.$$

To answer your question, the x value in the answer key was just a stand-in for an arbitrary real number. We could have used a or d instead of x to relate that bit of reasoning specifically to the problem, but I figured it would be easier to see just as a general statement on its own. As to the bit where I determined the values of a, b, c, and d, all I was doing was comparing the entries in the matrices in the equality

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} -a & -c \\ -b & -d \end{bmatrix}.$$

Since two matrices are equal when all of their entries are equal, this then tells us that $a = -a$, $d = -d$, $c = -b$, and $b = -c$. Solving these equations is straightforward: if $a = -a$, then $2a = 0$ so $a = 0$. By the same reasoning, $d = 0$. Since $c = -b$ and $b = -c$ are the same statement but with flipped signs, this just tells us that one of the variables is the negative of the other. So, our answer follows directly from this.

11. Lets say we have a change-of-basis matrix $P_{\mathcal{B}} = \begin{bmatrix} b_1 & \dots & b_n \end{bmatrix}$ such that

$$P_{\mathcal{B}}[x]_{\mathcal{B}} = x$$

Where the set $\{b_1, \dots, b_n\}$ form a basis for $\mathcal{B}$ and $[x] = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$ where $c_1, \dots, c_n$ are scalers such that $c_1 b_1 + \dots + c_n b_n = x$. Now, assume we know what both $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and what $P_{\mathcal{B}}$ are. How can we solve for $[x]_{\mathcal{B}}$ for these arbitrary sets/matrices? Is there a formula we can generalize to these arbitrary systems?
As you wrote, we know that

$$P_{\mathcal{B}}[x]_{\mathcal{B}} = x.$$

So, all we need to do to solve for $[x]_{\mathcal{B}}$ if you know both x and $P_{\mathcal{B}}$ is invert the matrix $P_{\mathcal{B}}$. That is, the following calculations get us where we want to go:

$$\begin{aligned} P_{\mathcal{B}}[x]_{\mathcal{B}} &= x \\ P_{\mathcal{B}}^{-1} P_{\mathcal{B}}[x]_{\mathcal{B}} &= P_{\mathcal{B}}^{-1} x \\ I[x]_{\mathcal{B}} &= P_{\mathcal{B}}^{-1} x \\ [x]_{\mathcal{B}} &= P_{\mathcal{B}}^{-1} x. \end{aligned}$$

So, it's nothing fancy. If you want to get rid of a square matrix that you know has linearly independent columns, just invert it!

12. I think I have two questions for review.

    (a) I was confused why you insisted in class for us to use $\dim\big(\text{Col}(A)\big)$ instead of $\text{rank}(A)$. Is there some sort of difference in definition between rank and $\dim\big(\text{Col}()\big)$, or are they equivocal? I keep thinking that there is some difference between the two because isn't by denoting something as dim(), doesn't that make it the maximum dimension. Or does dim() denote that simply the quantity of the dimension? Also, is rank(A) like $\dim\big(\text{Col}(A)\big)$, but transformed into $\mathbb{R}^n$, or transformed linearly at least. I read in the lecture notes that $\text{rank}(A) = \text{rank}(A^T)$, but $\text{Row}(A) = \text{Col}(A^T)$, so is rank like a metaphorical shadow of itself when transformed linearly? This probably sounds like rambling, but I hope what Im asking is making sense.
    You're overthinking it in this case, it's way simpler than any of this. The rank of a matrix A is *always, always, always* exactly the same as $\dim\big(\text{Col}(A)\big)$, and there's nothing more to it. The rank of a matrix is *just another way of saying the dimension of the column space*. They mean exactly the same thing *by definition*. That is, rank is just a "nickname" for the dimension of the column space. Nothing fancy, it's exactly the same type of thing as calling somebody "Tony" when their real name is "Anthony."

13. If you have an basis $\mathcal{B} = \{p_1, p_2, p_3\}$ where $p_1 = 1 - t$, $p_2 = 2 - t^2$, and $p_3 = 3t - 2t^2$, and you trying to find if its linearly dependent in $\mathbb{P}_2$., (like an example in class) how do you make the polynomials into vectors to solve this?
To make a set of polynomials (like the ones you have here) "into vectors" (a statement which is minorly misleading, since polynomials *are vectors* in $\mathbb{P}_2$), we need to choose a basis first.

A good, easy one to work with is the standard basis for $\mathbb{P}_2$, which is the set of vectors $\mathcal{E} = \{1, t, t^2\}$. So, the polynomials you have may be written, in $\mathcal{E}$-coordinates, as

$$[p_1]_{\mathcal{E}} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad [p_2]_{\mathcal{E}} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \quad [p_3]_{\mathcal{E}} = \begin{bmatrix} 0 \\ 3 \\ -2 \end{bmatrix}.$$

After this, showing that these vectors are linearly indpendent is easy: Just form a matrix whose columns are the vectors above and row-reduce.

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 3 \\ 0 & -1 & -2 \end{bmatrix} \overset{\text{Row Reduce}}{\Longrightarrow} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Notice that we get the identity, which means that the matrix is invertible. Since an $n \times n$ invertible matrix has n linearly independent columns which, together, span $\mathbb{R}^n$ (by the Invertible Matrix Theorem), this tells us that $\{[p_1]_{\mathcal{E}}, [p_2]_{\mathcal{E}}, [p_3]_{\mathcal{E}}\}$ is a linearly independent set of vectors which spans $\mathbb{R}^3$. As a set of vectors $\{v_1, \ldots, v_p\}$ belonging to a vector space V is linearly independent if and only if $\{[v_1]_{\mathcal{B}}, \ldots, [v_p]_{\mathcal{B}}\}$ is linearly independent for any basis $\mathcal{B}$ of V, it follows that $\{p_1, p_2, p_3\}$ is a linearly independent set in $\mathbb{P}_2$. Moreover, $\{p_1, p_2, p_3\}$ is actually a *basis* for $\mathbb{P}_2$, since it also spans $\mathbb{P}_2$!